

MANUSCRIPT DRAFT

Classical Economic Thermodynamics: The Fundamental Laws of Motion of Value and Social Reproduction

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Philip Mirowski argued that economics sought but lacked a genuine conservation law. This paper argues that classical political economy provides one — but the correct physics analogy is thermodynamics, not mechanics. The neoclassical programme fails because its only conservation is the tautological budget constraint (Hands 1993); the Sonnenschein–Mantel–Debreu theorem destroys any macro-level invariance. The classical tradition offers two genuine analogues. A *first-law* analogue: value is conserved in exchange, with the Monetary Equivalent of Labour Time (MELT) providing the aggregate conservation identity for any price system — a result proven in full generality by Shaikh (*Capitalism*, 2016). A *second-law* analogue: labour is an irreversible thermodynamic process, producing entropy while creating value. Abstract labour is the economic expression of this metabolic irreversibility; the MELT bridges thermodynamic substrate to monetary form. The Hamiltonian formalism of Goodwin and Flaschel–Semmler describes the first-law redistribution dynamics under simple reproduction; Shaikh’s three-layer macro dynamics trace the non-conservative regime of expanded reproduction toward the secular entropy maximum. The ecological crisis is the second law asserting itself at the system’s natural boundary.

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1 Introduction

Philip Mirowski's *More Heat than Light* (1989) posed a challenge that sits at the centre of political economy: can economics possess a genuine conservation law, analogous to energy conservation in physics? Mirowski traced the origins of both the classical and neoclassical research programmes to this founding ambition, and found both wanting. Neoclassical economics imported the energy metaphor without the substance, producing a formalism that conserved only accounting identities. Classical political economy left its invariant — labour-time — mathematically underspecified. The challenge has remained largely unanswered.

This paper argues that Mirowski's demand can be met — but only if the correct physics analogy is recognised as *thermodynamics*, not mechanics. The distinction is fundamental. Mirowski focused on classical mechanics: Hamiltonians, conservative force fields, potential energy. A mechanical analogy can only describe a *closed, reversible, self-sustaining* system — one that runs indefinitely without external input, like a frictionless pendulum or a planetary orbit. The capitalist economy is none of these things. It is open, irreversible, and requires the continuous expenditure of living labour to reproduce itself. Every hour of production consumes the worker's metabolic energy, transforms materials, and produces entropy that cannot be reclaimed. These are second-law processes, not mechanical ones.

The correct analogy therefore requires two thermodynamic principles operating at different levels. The **first law** (conservation): value is conserved in exchange — the monetary value of the net product is determined entirely by living labour, mediated by the Monetary Equivalent of Labour Time (MELT), and this total is neither created nor destroyed in circulation. The **second law** (entropy and irreversibility): labour is an irreversible thermodynamic transformation. Abstract labour — Marx's reduction of all concrete labours to their common substance — is the economic abstraction of this metabolic irreversibility. The MELT converts the thermodynamic quantity (hours of irreversibly expended human energy) into monetary value. Foley and Smith's Santa Fe lectures (2007) identified this thermodynamic structure but did not fully develop its implications for the conservation programme. The present paper does.

This challenge must be situated within the broader project of classical political economy. Duncan Foley (*The Unholy Trinity*) identifies a feature that distinguishes Smith, Ricardo, and Marx from the neoclassical programme that displaced them: the classical economists understood the economic system as an *ever-evolving, growing division of labour* in the form of increasingly complex and interdependent productive tasks. For them, the economy was not a static exchange mechanism among given endowments but a self-expanding system of social production. This historical and processual character is precisely what makes the thermodynamic analogy apt: what is conserved is an invariant of an evolving, entropy-producing, labour-driven system — not the static potential of a frictionless mechanism.

The paper proceeds as follows. Section 2 reconstructs Mirowski's critique of neoclassical economics, introducing the zeroth-law framing of the budget constraint and the Walrasian/Hicksian distinction. Section 3 presents the classical conservation principle through Smith, Ricardo, and Marx. Section 4 develops Shaikh's formal framework — labour values,

prices of production, and the MELT — as the first law of classical economic thermodynamics: value is conserved in exchange. Section 5 analyses the conservative dynamics under *simple reproduction* — the ideal-type first-law regime — introducing the Hamiltonian formally and deriving it from first principles, then examining redistribution across social classes (Goodwin), across processes of production (Flaschel–Semmler), and across the financial circuit (Foley), deriving the circuit Hamiltonian $\mathcal{H} = K/\mu$ and proving $\dot{\mathcal{H}} = 0$ under simple reproduction; the section closes with a taxonomy of frictions that break the conservative condition locally while preserving it on average. Section 6 develops the second law: labour as irreversible thermodynamic work, abstract labour as metabolic irreversibility, and the ecological dimension as the second law at the system’s natural boundary. Section 7 turns to *expanded reproduction*: §7.2 derives the unified $\dot{\mathcal{H}}$ formula across Goodwin, Flaschel–Semmler, and Foley, showing that all three yield $\dot{\mathcal{H}} = s_c \varepsilon L / (1 + \varepsilon)$ — the surplus labour time reinvested — as the thermodynamic intensity of accumulation; §7.3 develops Shaikh’s three-layer macro framework — Goodwin cycle, Juglar investment cycle, and Kondratiev long wave — tracing how each layer corresponds to a distinct pattern of Hamiltonian non-conservation and interpreting the secular decline in r^* as the system’s approach to its thermodynamic entropy maximum. The paper concludes that the ecological crisis is the second law asserting itself at the system’s natural boundary.

2 The Neoclassical Failure: Conservation as Tautology

Mirowski’s central historical argument is that the marginal revolution of the 1870s — Jevons, Walras, Menger — was a systematic transplantation of classical mechanics into economics, not the spontaneous discovery of rational choice theory. The objective was to achieve what Helmholtz, Joule, and Clausius had achieved in physics: a principle of invariance against which all transformations of the system could be measured. The neoclassical attempt located this invariant in utility.

Walras modelled exchange explicitly as force balance — marginal utility as generalised force, quantities as displacements, equilibrium as energy minimum. The differential relation $dU = \sum_i p_i dx_i$ implied a potential function with demand as its gradient. Slutsky symmetry guaranteed that compensated substitution effects formed a conservative vector field. Afriat’s Theorem (1967) gave this an empirical interpretation: the Generalised Axiom of Revealed Preference as a testable integrability condition.

Before locating the failure, it is important to identify what is shared. The budget constraint $p \cdot x = m$ — equivalently, Walras’s Law $p \cdot z(p) = 0$ for excess demand — is a *zeroth-law* condition that both traditions accept. In thermodynamics, the zeroth law establishes thermal equilibrium as a transitive relation, grounding the existence of temperature as a state variable; it is the precondition for any subsequent conservation law. The budget constraint plays the same role: it grounds the existence of prices as state variables, establishes the

transitivity of exchange, and is accepted by neoclassical and classical economics alike. The dispute is not about the zeroth law but about what the *first law* should be built upon it.

The neoclassical programme attempts to build a first law on utility. But as Hands (1993) clarified, this conservation holds only in *compensated* (Hicksian) demand space, not in observable Walrasian demand. The distinction is fundamental. **Walrasian (Marshallian) demand** $x^M(p, m)$ holds nominal income m fixed when prices change — this is the observable demand. **Hicksian (compensated) demand** $x^H(p, \bar{u})$ holds utility $\bar{u}$ fixed and adjusts income implicitly — this is *unobservable*, recoverable only by decomposing the observed Marshallian response via the Slutsky equation $\frac{\partial x_i^M}{\partial p_j} = \frac{\partial x_i^H}{\partial p_j} - x_j^M \frac{\partial x_i^M}{\partial m}$. The Slutsky substitution matrix for Hicksian demand is symmetric negative semidefinite, forming a genuine conservative vector field with the expenditure function $e(p, \bar{u}) = \min_x \{p \cdot x : u(x) \geq \bar{u}\}$ as its potential: $x_i^H = \partial e / \partial p_i$.

The conserved object is therefore the *expenditure function*, not utility. But the conservation of the expenditure function is an implication of the budget constraint supplemented by utility maximisation — it is contingent on the existence of a consistent preference ordering, not a discovery about the physical world. Afriat’s Theorem (1967) gives this an empirical handle: the Generalised Axiom of Revealed Preference (GARP) is the testable condition for whether an expenditure function consistent with utility maximisation could have generated the observed Marshallian data. Passing GARP confirms that a Hicksian conservative field *could* underlie the data — but GARP tests integrability, not the ontological claim that utility is energy. The neoclassical “conservation principle” thus reduces to: the budget constraint (zeroth law) plus preference consistency (a regularity assumption) yield a conservative expenditure field in unobservable compensated space. This is not energy conservation; it is a tautological restatement of budget feasibility dressed in potential-function language. A genuine conservation law is a contingent empirical claim about the world. The expenditure function is an artefact of the optimisation structure.

The integrability failure compounds at the aggregate level. The Sonnenschein–Mantel–Debreu theorem establishes that aggregate excess demand functions need not satisfy any regularity condition that applies to individual demand. Even if each consumer maximises a well-behaved utility function, their aggregate behaviour imposes no constraint on the shape of aggregate demand. Any continuous function satisfying Walras’s Law is admissible. No macro-level conservation survives. Friedman’s (1953) *as if* defence institutionalised the retreat: models became predictive instruments rather than representations of causal invariants, thereby abandoning the conservation programme entirely.

The Cambridge capital controversies (Cohen and Harcourt 2003) administered a further blow from within formal economics. The neoclassical theory of distribution requires that capital be measurable in physical units independent of distribution — so that the marginal product of capital can determine the rate of profit. But the value of heterogeneous capital goods depends on the rate of profit, which it is supposed to explain. This circularity (the Wicksell effect) generates the anomalies of reswitching and capital-reversing, acknowledged by Samuelson (1966) as logical necessities. There is no consistent aggregate capital measure. The neoclassical programme cannot even specify the invariant it would need to conserve.

The CCC also clarify the precise relationship between the neoclassical and classical traditions. The neoclassical aggregate production function $Y = F(K, L)$ is valid — and the marginal product $\partial F/\partial K$ equals the profit rate without circularity — if and only if the value composition of capital $\Omega_j = C_j/v_j$ is uniform across all sectors. Under this condition, relative prices equal relative labour values exactly (the Wicksell effect vanishes), and capital K aggregates consistently. The neoclassical production function is therefore the *special case* of the labour theory of value in which all production processes have equal profit-wage ratios — Ricardo’s “93 per cent” approximation becomes exact. Reswitching demonstrates this special case is structurally non-generic: as soon as Ω_j differs across sectors, the aggregate production function fails, and the classical framework (prices of production, Shaikh’s universal conservation identity) is the correct general treatment.

One further observation deserves emphasis. The neoclassical programme never engaged with the *second law* at all. Utility theory has no concept of irreversibility: a consumer who maximises utility over time could, in principle, reverse every decision without loss. Production in neoclassical theory is a reversible transformation of inputs into outputs, governed by production functions that are symmetric in time. There is no entropy, no metabolic cost, no thermodynamic limit. The second law is simply absent from the neoclassical research programme. This absence is not a detail but a fundamental failure: an economics without irreversibility has no theory of why the past is different from the future, why production requires labour, or why ecological limits exist.

3 The Classical Principle: Value Conserved in Exchange, Created in Production

The classical tradition’s engagement with conservation begins with Adam Smith’s *Wealth of Nations* (1776). Smith’s dual account of value — labour *commanded* (how much labour a commodity can purchase) and labour *embodied* (how much went into producing it) — contains, despite its inconsistency, the germ of the conservation principle. In the labour-embodied reading, the price at which a commodity exchanges is regulated in the long run by the labour required for its production. Exchange redistributes the product of labour; it does not create value. The “real price” of everything is, in the last analysis, the toil and trouble of its production.

David Ricardo sharpened this into a systematic claim. In the *Principles* (1817), relative exchange values are regulated by relative labour times, and changes in the wage-profit distribution leave them approximately unchanged — his famous “93 per cent” claim. But Ricardo was honest enough to perceive the problem his own formulation created: a rise in wages alters relative prices differentially, depending on the capital intensity of each sector, so that no produced commodity can serve as an invariable measure of value immune to distributional

changes. Ricardo died in 1823 believing this problem insoluble, leaving what he called “the most difficult subject in Political Economy” to his successors.

Karl Marx resolved the confusion between concrete and abstract labour that had trapped Ricardo. Concrete labour is the specific activity performed — weaving, smelting, teaching. Abstract labour is what all concrete labours have in common: the expenditure of human labour-power, stripped of its specific useful form, validated in exchange as socially necessary. It is abstract labour that constitutes the substance of value.

The conservation principle follows with precision. In exchange, commodities are equated as values: both contain the same quantity of abstract labour. Exchange creates no value; it only realises it. Shaikh (1977, p. 115) states this formally as a theorem: “no Value is created in the circulation process” — citing Marx’s own formulation in *Capital* (Vol. II, Ch. VI): “if the commodities are not sold at their values, then the sum of converted values remains unchanged; the plus on one side is a minus on the other.” Whatever prices prevail, the total value distributed in exchange is fixed by what was created in production.

Production is the site of value *creation*. Capital purchases means of production and labour-power and sets in motion a process that returns more value than was advanced. Workers produce, in the first part of the working day, the equivalent of their wage (necessary labour); they continue beyond this point (surplus labour), producing surplus value appropriated by capital. The conservation and exploitation principles are inseparable: value is conserved in exchange *precisely because* new value is produced in production by surplus labour, and the surplus is extracted rather than conjured from nowhere. Any transfer in exchange is redistributive; any genuine increment is productive.

The classical conservation principle: the total value of the net product is conserved in exchange (redistributed, not created or destroyed) and created in production by living labour alone. This is the first-law claim of classical political economy. Its full mathematical specification requires the formal framework of the next section.

4 First Law of Classical Economic Thermodynamics: Value is Conserved in Exchange

4.1 Price Decomposition and Vertically Integrated Values

The formal starting point is the decomposition of any commodity price into its component parts. Following Shaikh’s framework as presented by Işıkara and Mokre (2026, Ch. 3), for any sector j :

$$p_j = u_j + \pi_j + m_j \tag{2.1}$$

where $u_j = w_j(L_j/X_j)$ is unit wage cost, π_j is unit gross profit, and $m_j = \sum_i p_i a_{ij}$ is unit material cost. This decomposition is recursive: each material input decomposes in turn into wage, profit, and further material components. Substituting recursively resolves any price into an infinite sum of wage and profit components traced through all direct and indirect inputs — exactly what the Leontief inverse captures in closed form.

To compare labour across sectors with heterogeneous wages — because concrete labours must be reduced to a common unit of abstract labour — Shaikh introduces skill-adjusted labour coefficients:

$$gl_j = \frac{1}{\bar{w}} \frac{W_j}{X_j} = \frac{w_j}{\bar{w}} \frac{L_j}{X_j} \quad (3.1)$$

where $\bar{w}$ is the economy-wide average wage. The ratio $w_j/\bar{w}$ weights each sector's hours by relative wages — the market's own valuation of the relative productivity of different concrete labours, following Marx's reduction of skilled to unskilled labour through social validation. Total labour values — the sum of all direct and indirect abstract labour required for a unit of output, including depreciation of fixed capital — are then:

$$v = gl(I - A - D)^{-1} \quad (3.2)$$

where D is the depreciation matrix and $(I - A - D)^{-1}$ is the extended Leontief inverse. This is the complete vertically integrated labour content: the full genealogy of abstract labour expenditure, from raw materials through all stages of production.

4.2 Prices of Production and Sraffa's Standard Commodity

Prices of production — prices that equalise profit rates across sectors in competition — are given by:

$$p^* = (1 - r/R) v (I - Rr/R H)^{-1} \quad (3.3)$$

where r is the actual profit rate, R the maximum rate of profit (at zero wages), and H a matrix encoding the distribution of capital intensities (İşıkara and Mokre 2026, Ch. 3; Shaikh 1984). At $r = 0$: p^* equals labour values v . As r rises toward R , prices deviate from values in a systematic pattern determined by organic composition. The deviation is not random but structured: sectors with above-average capital intensity gain relative to their labour values; sectors with below-average intensity lose. Competition redistributes surplus value among sectors via this price mechanism.

The matrix H is intimately related to Sraffa's *standard commodity* — the composite good in which every industry appears in the same proportions as the economy as a whole. The standard commodity is the leading eigenvector of A : if q satisfies $Aq = q/(1 + R)$, then q is

the standard commodity and $R = (1/\lambda_1) - 1$ (where λ_1 is the Perron-Frobenius dominant eigenvalue). In the standard system, the wage-profit trade-off is exactly linear:

$$r = R \left(1 - \frac{w}{w_{\max}} \right)$$

which means the price of the standard commodity is immune to changes in the distribution of income. This is Ricardo’s invariable measure of value, constructed exactly. Equation (3.3) shows that p^* reduces to labour values at $r = 0$ and diverges at $r = R$: the classical tradition’s claim that labour values govern prices is exact at the competitive floor and empirically confirmed — with 10% mean deviations — by Işıkara and Mokre (2021) across 42 countries.

4.3 The MELT: Conservation Identity and the Exploitation Rate

The Monetary Equivalent of Labour Time bridges the abstract world of labour-time values and observable monetary prices:

$$\mu = \frac{p^\top (I - A)x}{\ell^\top x} \quad (\text{MELT})$$

where $p^\top (I - A)x$ is the monetary value of the net product (national income) and $\ell^\top x = L$ is total living labour in hours. The aggregate conservation identity follows:

$$\boxed{p^\top (I - A)x = \mu L} \quad (\text{Conservation})$$

Shaikh (*Capitalism*, 2016, Ch. 6, §IV) proves this in its most general form: aggregate money profit equals aggregate surplus value for *any* relative price system, not just prices of production:

$$\Pi = S = (\mu - w)L \quad \text{for any price system.} \quad (\text{Universal})$$

Individual sectors transfer value to each other via price deviations — sectors with above-average organic composition gain, those below lose — but these transfers are exactly zero-sum at the aggregate. The “transformation” from values to prices is a redistribution within a conserved total, not a violation of conservation. This is Shaikh’s resolution of what he calls the “Universal Transformation Problem.”

The relationship between the MELT and the exploitation rate follows directly. Total new value is μL ; workers receive wages wL ; surplus is $(\mu - w)L$:

$$\boxed{\varepsilon = \frac{\mu}{w} - 1 \quad \Longleftrightarrow \quad \mu = w(1 + \varepsilon)} \quad (\text{Exploitation})$$

The MELT equals the money wage times one plus the exploitation rate. This single equation encapsulates the class relation: productivity growth (rising μ) at constant wages raises exploitation; wage growth at constant μ reduces it. The MELT is empirically measurable as money GDP divided by total hours of productive labour — making the exploitation rate an observable quantity, not merely a theoretical category.

4.4 What Is Conserved: Shaikh’s 1977 Proof

The deepest statement of the conservation principle appears in Shaikh’s (1977) “Marx’s Theory of Value and the ‘Transformation Problem.’” In Section II.6 (“Production and Circulation”), Shaikh, following Marx (*Grundrisse*, Notebook VI), writes: “no Value is created in the circulation process.” The consequences are exact: “regardless of the actual money prices at which these commodities are sold, only the same mass of commodities (and hence the same amount of Value) exists after the sales as before.” He cites Marx directly (*Capital*, Vol. II, Ch. VI):

If commodities are sold at their values, then the magnitude of value in the hands of the buyer and seller remains unchanged. Only the form of existence of value is changed. If the commodities are not sold at their values, then the sum of converted values remains unchanged; the plus on one side is a minus on the other.

The conserved quantity is the total sum of Values = $C + L$ (constant capital transferred plus living labour added), decomposing as $C + V + S$. Any deviation of prices from values is zero-sum: it redistributes value without creating or destroying it.

The specifically capitalist conservation identity is the relationship between aggregate surplus value and aggregate profit. Shaikh establishes it in two steps. First, the value rate of profit in sector k is:

$$\rho_k = \frac{S_k}{C_k + V_k} = \frac{S_k/V_k}{(C_k + V_k)/V_k} = \frac{\varepsilon_k}{1 + \Omega_k}$$

where $\varepsilon_k = S_k/V_k$ is the sectoral exploitation rate and $\Omega_k = C_k/V_k$ is the **value composition of capital** (constant to variable capital). The ratio of exploitation rate to value composition determines the sectoral profit rate. Since the rate of surplus value is equalized across sectors (all workers face the same working day and social wage structure), differences in ρ_k arise entirely from differences in Ω_k . Note that $\Omega_k = C_k/V_k$ is distinct from the capital-labour ratio $\kappa = C/l$ used in the long-wave formula: they are related by $\Omega = \kappa/w$ (the wage rate converts between labour-time and variable-capital units). Second, competition equalises money rates of profit by deviating prices from values. But any relative price movement that raises sector i ’s price relative to sector j ’s is simultaneously a redistribution — the gain on one side is a loss on the other. The aggregate money profit therefore remains equal to aggregate surplus value:

$$\Pi = \sum_k \Pi_k = S = L - V = (\mu - w)L \quad (\text{S77.1})$$

This is Shaikh’s conservation invariant: aggregate money profit equals aggregate surplus value, regardless of the particular relative prices prevailing. The transformation from direct prices to prices of production is a transformation of the *distribution* of profit, not of its *total*. In Shaikh’s iterative procedure, Marx’s transformation is the first step of an iterative process (direct prices $\rightarrow$ prices of production $\rightarrow \dots \rightarrow$ convergence); at each step, relative prices change and value transfers between sectors, but (S77.1) holds throughout. The aggregate is fixed at every stage.

4.5 The Universal Transformation Problem: *Capitalism* Chapter 6

Chapter 6 of Shaikh’s *Capitalism* (2016), “Capital and Profit,” extends this into a fully general treatment. Section II identifies two apparent sources of aggregate profit: (i) surplus labour, and (ii) transfers of value via relative price changes. The chapter then demonstrates that (ii) is zero-sum at the aggregate level — the *only* net source of aggregate profit is surplus labour. Section III states this as a theorem: “No aggregate profit without surplus labour”; “Positive profits require surplus labour.”

Section IV, “Aggregate Profits and Transfers of Value: a General Solution to the Universal ‘Transformation Problem,’ ” makes the conservation explicit in its most general form. The key result is equation (S2016.1):

$$\Pi = S \quad \text{for any relative price system}$$

— not just for prices proportional to values and not just for prices of production, but for *any* prices whatsoever. This is the deepest expression of the conservation principle: aggregate profit equals aggregate surplus value unconditionally. Subsection IV.1 shows that relative price changes transfer value between sectors but sum to zero over the whole economy. Subsection IV.2 examines how the composition of output affects the distribution of value transfers, showing that this too leaves the aggregate unchanged.

The significance for Mirowski’s challenge is direct. Shaikh’s Ch. 6 demonstrates that value is conserved at the aggregate level in precisely the sense Mirowski demanded: a contingent empirical claim, not a definitional tautology, linking living labour to monetary profit across all possible price systems. The MELT μ makes this operational — $\Pi = (\mu - w)L$ holds for any p because μ is defined to ensure it holds, and the empirical content is the stability and measurability of μ across time and national accounts. Işıkara and Mokre’s (2021) finding of bounded, systematic price-value deviations across 42 countries is the empirical confirmation: the gravitational pull of values on prices is real, and the conservation holds in the aggregate even as individual sector prices deviate in predictable directions.

5 Simple Reproduction: Conservative Dynamics

Under simple reproduction, living labour in each period exactly replaces the consumed means of production and reproduces workers' subsistence. The economy returns to the same scale at the end of each production cycle: no value accumulates across periods. This is the ideal-type case for the first law in its pure form. A given total μL circulates among sectors and classes without net creation or destruction.

5.1 The Hamiltonian: Definition and Economic Meaning

The analysis of simple reproduction requires a mathematical tool — the **Hamiltonian** — that will be unfamiliar to readers outside mathematical physics. It deserves a careful introduction before it is deployed.

In classical mechanics, the Hamiltonian $H(q, p)$ is a scalar function of *generalised coordinates* q (position-like variables, describing the configuration of the system) and *conjugate momenta* p (velocity-like variables, describing the rate of change). It generates the complete equations of motion via **Hamilton's equations**:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

The key property: when H has no explicit time dependence ($\partial H / \partial t = 0$), total differentiation along any trajectory gives

$$\dot{H} = \sum_i \left(\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) = \sum_i (-\dot{p}_i \dot{q}_i + \dot{q}_i \dot{p}_i) = 0.$$

H is therefore a **conserved quantity** — a constant of motion along every trajectory of the system. In mechanics H is total energy (kinetic plus potential); its conservation is the first law of mechanics.

In the economic context developed here, H measures the total *distributional tension* of the system — how far the economy departs from its long-period position of equalised profit rates and balanced sectoral proportions. When $\dot{H} = 0$, this tension neither increases nor decreases: the system cycles through distributional states in a closed orbit, always returning to the same aggregate. Value is redistributed between wages and profits, between sectors, between debtors and creditors — but the total is conserved. When $\dot{H} \neq 0$, the orbit is not closed: the system is gaining or losing “distributional energy,” which corresponds to net creation (production) or net destruction (crisis) of value.

The three conditions under which simple reproduction admits a Hamiltonian with $\dot{H} = 0$ are precisely the conditions that allow the economic dynamics to be written in the form of Hamilton's equations:

1. **Fixed technique** (A, ℓ, μ constant): $\partial H/\partial t = 0$ — the energy landscape is time-invariant, so no structural drift enters the conservation equation.
2. **Constant scale** ($\dot{K} = 0$): no new value is injected — the “total energy” μL is fixed each period. Without this, a source term enters and $\dot{H} > 0$ by construction.
3. **Frictionless competition** ($\Delta_p = \delta_p I, \Delta_x = \delta_x I$): adjustment dynamics are *symplectic* — they preserve volumes in (q, p) phase space. Unequal adjustment speeds across sectors introduce dissipation, breaking the symplectic structure and causing $\dot{H} \neq 0$.

Under these three conditions, the competitive redistribution of surplus value traces a closed Hamiltonian orbit. The following two subsections derive the explicit Hamiltonian for redistribution across social classes (Goodwin) and across processes of production (Flaschel–Semmler), and the third shows how relaxing condition 3 — via financial dynamics — introduces the first departure from exact conservation.

5.2 Redistribution Across Classes: Goodwin’s Growth Cycle

Goodwin’s (1967) predator-prey model couples the employment rate λ and wage share ω :

$$\dot{\lambda} = \lambda(a - b\omega), \quad \dot{\omega} = \omega(-c + d\lambda).$$

To derive the Hamiltonian, substitute $q = \ln \lambda, p = \ln \omega$, so that $\dot{q} = \dot{\lambda}/\lambda = a - be^p$ and $\dot{p} = \dot{\omega}/\omega = -c + de^q$. These are Hamilton’s equations — $\dot{q} = \partial H/\partial p$ and $\dot{p} = -\partial H/\partial q$ — for:

$$H(q, p) = ap - be^p + cq - de^q, \quad \dot{H} = 0.$$

One verifies directly: $\partial H/\partial p = a - be^p = \dot{q}$ and $-\partial H/\partial q = -(c - de^q) = -c + de^q = \dot{p}$. The three conditions of simple reproduction ensure each step holds: fixed technique gives constant a, b, c, d ; constant scale prevents $\dot{K}$ from entering $\dot{q}$; frictionless competition means the Lotka-Volterra structure is exact rather than approximate.

The conservation $\dot{H} = 0$ has a precise political-economic meaning. The Hamiltonian H measures the distributional tension — how far the wage share and employment are from their classical equilibrium values $(\lambda^*, \omega^*) = (c/d, a/b)$. $\dot{H} = 0$ means this tension is conserved: the class struggle over distribution traces a closed orbit in which neither capital nor labour wins decisively. As wages rise (workers exercise bargaining power at high employment), profits compress, accumulation slows, employment falls, and profitability is restored. The cycle returns to its starting distribution. No value is created or destroyed — only the division of the existing total changes.

Hamilton’s equations make the mechanical analogy explicit and exact. In classical mechanics, q is position, p is momentum, $\partial H/\partial p = \dot{q}$ is velocity, and $-\partial H/\partial q = \dot{p}$ is the restoring force. The Goodwin system maps onto this structure term by term:

- **Position** $q = \ln \lambda$: where the economy stands in employment space — how deployed labour is in the circuit of capital. Employment is the coordinate of capital's absorption of living labour.
- **Momentum** $p = \ln \omega$: the accumulated distributional power of the working class — how much of the value product has been claimed as wages. Just as mechanical momentum is the inertia of motion, wage share is the inertia of workers' distributional gains, built up at high employment and eroded by the reserve army.
- **Velocity** $\partial H / \partial p = a - b\omega = \dot{q}$: the net rate of employment growth — how fast the economy moves through employment space. High wage share ω slows accumulation and hence employment growth; low ω accelerates it. This is the accumulation complement of the Phillips curve: wage pressure dampens the velocity of capital deployment.
- **Restoring force** $-\partial H / \partial q = -c + d\lambda = \dot{p}$: the net wage-share acceleration — how fast workers' distributional momentum is building or decaying. At high employment $\lambda > c/d$, workers have bargaining power and $\dot{p} > 0$ (the wage share is pushed upward); at low employment, the reserve army exerts downward pressure and $\dot{p} < 0$. This is the Phillips curve: employment is the force driving wage-share momentum.

The Hamiltonian decomposes accordingly:

$$H = \underbrace{(ap - be^p)}_{\text{wage-momentum energy}} + \underbrace{(cq - de^q)}_{\text{employment-position energy}}$$

The first term is the *wage-momentum energy* — the contribution of workers' distributional position to total class tension; maximised at $\omega = \omega^*$ and falling away on both sides. The second is the *employment-position energy* — the contribution of the labour market position to the tension; maximised at $\lambda = \lambda^*$. The total H measures how far the system is from the distributional truce in both dimensions simultaneously. It is the precise thermodynamic measure of **the intensity of class struggle**: the amplitude of the orbit in the (λ, ω) plane, conserved over the cycle because neither side wins decisively.

The exploitation rate $\varepsilon = \mu/w - 1$ enters the equilibrium directly: $\varepsilon^* = (1 - \omega^*)/\omega^*$. Neither wages policy alone nor profit-sharing alone can stabilise the cycle, because the oscillation is structural — built into the reproduction circuit by the three conditions above.

5.3 Redistribution Across Processes: Flaschel–Semmler

Flaschel and Semmler (1987) extend the Hamiltonian structure to an n -sector economy. Let $M = B - RA$ where B is the output matrix, A the input matrix, and R a uniform profit factor. With $y = \ln p$ and $z = \ln x$ as the price and quantity coordinates, price and quantity adjustments follow:

$$\dot{y} = -\Delta_p M e^z, \quad \dot{z} = \Delta_x M^\top e^y.$$

These are Hamilton's equations for (z, y) when $\Delta_p = \delta_p I$ and $\Delta_x = \delta_x I$ (condition 3), with Hamiltonian:

$$H(z, y) = \delta_x \mathbf{1}^\top M^\top e^y + \delta_p \mathbf{1}^\top M e^z, \quad \dot{H} = 0.$$

The mechanical analogy is equally precise at the sectoral level. The same four-fold structure — position, momentum, velocity, force — maps onto competitive dynamics sector by sector:

- **Position** $z_k = \ln x_k$: where sector k stands in output space — how far its production is from the Sraffian equilibrium proportion. Output is the coordinate of capital's deployment across industries.
- **Momentum** $y_k = \ln p_k$: the price “momentum” of sector k — how far its price is above or below the cost-covering level. Price is the accumulated profitability signal: a sustained above-cost price is the inertia of competitive advantage, driving capital inflows.
- **Velocity** $\partial H / \partial y_k$: the rate at which sector k 's output adjusts in response to the profitability signal — the classical law of excess profitability. When price exceeds cost, new investment flows in and output rises; when below, capital exits and output contracts.
- **Restoring force** $-\partial H / \partial z_k$: the excess-demand pressure driving prices — the Walrasian law of excess demand. When output is below equilibrium, demand exceeds supply and the price is pushed up; above equilibrium, oversupply compresses the price back toward cost.

The Hamiltonian decomposes as:

$$H = \underbrace{\delta_x \mathbf{1}^\top M^\top e^y}_{\text{price-momentum energy (kinetic)}} + \underbrace{\delta_p \mathbf{1}^\top M e^z}_{\text{quantity-position energy (potential)}}$$

The first term — *kinetic value* — measures the aggregate profitability deviation: how far current prices are from cost-covering across all sectors. This is the kinetic energy of capital in competitive motion, seeking higher-profit destinations. The second — *potential value* — measures how far sectoral quantities are from Sraffian equilibrium proportions: the potential energy of quantity imbalances that drives prices back toward cost-covering. $\dot{H} = 0$ means these two forms of distributional energy convert into each other continuously in a closed orbit — competitive gravitation without net creation or destruction.

The equilibrium is the Sraffian long-period position: prices at p^* , quantities at standard commodity proportions. The standard commodity — leading eigenvector of A and Ricardo's invariable measure of value — is the gravitational centre of the value field. The key contrast with Goodwin is the level of analysis: **Goodwin operates at the class level** (labour vs. capital in aggregate, wage share vs. employment rate), while **FS operates at the sectoral level** (capital vs. capital across industries, prices vs. quantities). Goodwin H is the intensity of class struggle; FS H is the intensity of inter-sectoral competitive tension. Both are first-law invariants — redistribution without creation — but the orbit in Goodwin

traces the wage-profit cycle, while the orbit in FS traces the price-quantity competitive cycle across the input-output network.

When condition 3 fails — when $\Delta_p \neq \delta_p I$ across sectors — the system becomes non-conservative:

$$\dot{H} = \sum_i \Pi_i^{(x)} - \sum_i \Pi_i^{(p)} \neq 0,$$

where the Π terms measure sectoral power flows. Financial distortions, monopoly pricing, and supply-chain asymmetries enter as non-uniform adjustment speeds, introducing sources and sinks into the value field. This is the first bridge to expanded reproduction at the sectoral level.

5.4 The Hamiltonian Structure of the Circuit: Foley's Circuit of Capital

Foley's (1982, 1986) complete circuit-of-capital formulation reveals the Hamiltonian structure of the financial sphere. The circuit runs $M \rightarrow C \cdots P \cdots C' \rightarrow M'$: money capital M purchases commodities and labour-power, which enter production, yield new commodities C' , realised as money with profit $M' = M(1 + \pi)$. Under simple reproduction, the circuit closes exactly each period: $M' - M = \Pi = (\mu - w)L$, and the MELT conservation identity holds.

The circuit consists of three **stocks of value** — productive capital $N(t)$, commercial capital $X(t)$, and financial capital $F(t)$ — driven by capital outlays $C(t)$, value of finished product $P(t) = C(t - T_P)$, and sales $S(t) = (1 + q)P(t - T_R)$, where q is the markup on costs and T_P, T_R, T_F are the production, realisation, and financial lags. Sales decompose into cost-recovery $S'(t) = S/(1 + q)$ and surplus value $S''(t) = qS/(1 + q)$; fraction p of surplus is recapitalised. The three stocks evolve as:

$$\dot{N} = C - P, \quad \dot{X} = P - S', \quad \dot{F} = S - (1 - p)S'' - C$$

Summing directly:

$$\dot{K} \equiv \dot{N} + \dot{X} + \dot{F} = pS''$$

Capital grows if and only if $p > 0$. The MELT provides the thermodynamic bridge: $S'' = \mu S_\ell$, where $S_\ell = \varepsilon L/(1 + \varepsilon)$ is surplus labour time. Define the **circuit Hamiltonian** $\mathcal{H} \equiv K/\mu$ — total capital in abstract-labour-time units:

$$\boxed{\dot{\mathcal{H}} = \frac{\dot{K}}{\mu} = p S_\ell = \frac{p \varepsilon}{1 + \varepsilon} L}$$

In **simple reproduction** ($p = 0$): $\dot{\mathcal{H}} = 0$ — the circuit closes exactly, the Hamiltonian is conserved. The exploitation rate ε may be positive (surplus is extracted) but all of it is consumed rather than accumulated; the total capital stock K is unchanged; the first-law conservation holds in its pure form. The limit $\varepsilon \rightarrow 0$ gives $\dot{\mathcal{H}} = 0$ for any p : without surplus labour, no accumulation is possible and simple reproduction is the only option. The expanded reproduction case ($p > 0$) is taken up in Section 7.

The mechanical analogy for the circuit differs from Goodwin and FS: rather than a two-dimensional phase-space Hamiltonian with canonical (q, p) coordinates, $\mathcal{H} = K/\mu$ is the conserved scalar of the circuit’s aggregate dynamics. Its economic interpretation maps directly onto the mechanical concept of total energy:

- $\mathcal{H} = K/\mu$ is the total capital stock in abstract labour-time units — the economy’s capacity to command living labour. Just as mechanical energy measures a system’s capacity to do physical work, $\mathcal{H}$ measures capital’s claim on future labour: how many hours of living labour the accumulated capital stock represents at the current MELT.
- $\dot{\mathcal{H}} = 0$ (simple reproduction): capital’s claim on living labour is exactly conserved each period. Surplus is extracted and wages are paid, but the total stock returns to the same scale. The first law holds in pure form: value circulates without expansion.
- $\dot{\mathcal{H}} = pS_\ell$ is the economic analogue of power (energy per unit time): the rate at which additional labour-hours of capital are locked in through reinvestment of surplus. It is the rate at which the second law — irreversible labour expenditure — drives the first-law conservative manifold outward.

The three stocks N, X, F are the circuit’s three degrees of freedom — productive, commercial, and financial capital. The circuit “velocity” is the turnover rate $1/(T_P + T_R + T_F)$: how fast value completes the full loop from money to production to sale and back to money. The “force” driving the circuit is the profit rate $r = q/(T_P + T_R + T_F)$: the markup on costs that makes the circuit self-sustaining and, when $p > 0$, self-expanding.

5.5 Non-Conservative, Dissipative Frictions: Liquidity-Profit Cycles

The three conditions for $\dot{H} = 0$ — fixed technique, constant scale, frictionless competition — define an ideal. Real capitalist economies operate with frictions that break the third condition while preserving conservation on average. These frictions are violations of the $\Delta_p = \delta_p I$ requirement: they introduce asymmetries into the adjustment process that locally break the symplectic structure of the dynamics, without displacing the long-run attractor.

The **liquidity-profit rate cycle** (Foley 1987) is the canonical case. Firms hold a fraction λ of total wealth in liquid (monetary) form rather than investing it productively. The profit rate π and liquidity preference λ generate:

$$\dot{\pi} = \alpha \pi (\bar{u} - u(\pi, \lambda)), \quad \dot{\lambda} = \beta (\pi - r_\lambda)$$

where r_λ is the return on liquid assets. At equilibrium, the Jacobian has purely imaginary eigenvalues — closed orbits, as in the Goodwin and FS cases. When credit expansion reduces r_λ , a **Hopf bifurcation** occurs: a limit cycle emerges in the (π, λ) plane, locally $\dot{H} \neq 0$, but $\oint \dot{H} dt = 0$ over the full cycle. Credit creation temporarily injects monetary claims in excess of μL — the MELT identity is locally violated — and the subsequent financial contraction restores it by destroying the excess claims.

Taking log-coordinates $q = \ln \lambda$ (log liquidity ratio) and $p = \ln \pi$ (log profit rate), the mechanical analogy reads term by term:

- **Position** $q = \ln \lambda$: where capital stands on the spectrum from fully liquid (monetary, unproductive) to fully deployed in production. High λ means capital is sheltering in liquid form; low λ means it is engaged in the circuit.
- **Momentum** $p = \ln \pi$: the profit rate — the accumulated competitive pressure on capital. High π drives capital out of liquid positions into productive deployment; low π pushes it toward the safety of liquid assets.
- **Velocity** $\dot{q}$: the rate at which capital migrates between liquid and productive form, driven by the gap $\pi - r_\lambda$ between productive and liquid returns. When $\pi > r_\lambda$, firms shift out of liquid assets and λ falls — capital accelerates into the productive circuit.
- **Restoring force** $\dot{p}$: how fast the profit rate changes in response to capacity utilisation — the business-cycle mechanism by which overheating compresses profits and slack restores them.

The conserved Hamiltonian $H(\pi, \lambda)$ — which exists at the equilibrium where the Jacobian has purely imaginary eigenvalues — is the **intensity of financial tension**: how far the monetary circuit is from its equilibrium (π^*, λ^*) . $\dot{H} = 0$ on the ideal conservative orbit means the financial cycle neither amplifies nor decays: the orbit is closed and the total financial tension is preserved. When credit expansion shifts r_λ downward, the Hopf bifurcation breaks this exact conservation locally — the orbit opens into a limit cycle on which $\dot{H}$ oscillates around zero, restoring $\oint \dot{H} dt = 0$ only over the full financial period. The MELT identity is the thermodynamic anchor: any monetary claim in excess of μL is temporary, and the contraction that destroys it is the first law reasserting itself after a local violation.

Three further frictions operate by the same mechanism:

- **Markup pricing**: oligopolistic price-setting introduces a wedge between unit costs and revenues that dampens the cross-dual oscillation — condition 3 is broken by price stickiness rather than liquidity preference
- **Non-uniform capital mobility**: barriers to entry and exit mean sectors adjust profit rates at different speeds, breaking the $\Delta_p = \delta_p I$ uniformity condition
- **Wage rigidities**: imperfect real-wage flexibility introduces friction into the employment–wage–share orbit, causing the Goodwin orbit to spiral inward toward a stable node rather than tracing a closed curve

Each friction preserves the long-run attractor (prices of production, standard commodity proportions) and restores conservation on average over a full cycle. They are deviations from

the ideal conservative manifold — the realistic texture of circulation dynamics — rather than violations of the first law itself.

6 Second Law of Classical Economic Thermodynamics: Value is Created Only in Production

6.1 Why Mechanics Is the Wrong Analogy

Mirowski's critique was directed at the mechanics analogy. He showed that neoclassical economics failed to construct a genuine Hamiltonian system — it had no conserved invariant at the aggregate level. The natural response, which this paper follows, is to ask what the *correct* physical analogy is. The answer is not mechanics but thermodynamics — and specifically, both laws of thermodynamics, not just the first.

A mechanical system — a pendulum, a planetary orbit — is closed and time-reversible. Run the film backwards and the physics is unchanged. The Hamiltonian H is conserved because no energy enters or leaves, and no process produces entropy. The system requires no external input to sustain itself.

The economy is none of this. Film the economy backwards and it is absurd: workers become younger, factories reassemble from their products, ore reconstitutes from steel. The economy has a direction in time because production is irreversible. The second law rules the production sphere. Mechanics can at best describe the circulation of already-produced value — the first-law dynamics of redistribution. It cannot describe the production process itself.

6.2 Labour as Irreversible Thermodynamic Process

Every act of production involves three irreversible thermodynamic processes running simultaneously:

Metabolic expenditure. The worker burns calories, depletes ATP, fatigues muscles and nerves. This is a one-way thermodynamic process: chemical potential energy (food) is converted to mechanical and cognitive work and then to heat. The worker cannot un-expend the metabolic energy. Rest and food restore labour-power, but at thermodynamic cost — more entropy produced in the reproduction of the worker than was produced in the work itself.

Material transformation. Ore becomes steel, cotton becomes cloth, silicon becomes a chip. These transformations are irreversible: the second law forbids spontaneous reassembly. The entropy of the material system increases with every productive transformation.

Temporal expenditure. The working day consumed is gone. Labour-time is the most irreversible resource in the economy: it cannot be stored, recovered, or reversed. The MELT

μ is a *flow* concept — value per unit time — precisely because time is the dimension along which labour is irreversibly expended.

The commodity produced is a local pocket of order — a structured, useful object with low internal entropy. But this local order is paid for by a greater increase in disorder elsewhere: in the worker’s metabolic system, in the depleted raw materials, in the waste heat of production. This is the second law: local entropy decrease (the ordered commodity) is always accompanied by global entropy increase (metabolic cost + material degradation + thermal dissipation). The commodity’s value is the social accounting of the total irreversible thermodynamic cost of its production.

6.3 Abstract Labour as Thermodynamic Quantity

Marx’s concept of abstract labour acquires its deepest interpretation here. The distinction between concrete and abstract labour is, read thermodynamically, the distinction between the specific thermodynamic process and the abstract quantity of entropy production it involves:

Concrete labour = the specific irreversible process: the particular chemical reactions, mechanical motions, and cognitive operations of weaving, smelting, programming. Each has its own thermodynamic profile.

Abstract labour = the homogeneous quantum of human metabolic energy expenditure, stripped of its specific form. What weaving and smelting and programming share is that they all consume human bioenergy irreversibly, at a rate measurable in time.

The social reduction of concrete to abstract labour — the process by which the market treats all labours as qualitatively identical and quantitatively comparable — is therefore the social recognition that all production involves the same kind of thermodynamic process: the irreversible expenditure of human metabolic energy. The MELT μ is the conversion factor between this thermodynamic quantity (hours of abstract labour expended) and its monetary expression. Like Boltzmann’s constant k_B , which converts between temperature (macroscopic) and mean molecular kinetic energy (microscopic), the MELT converts between the thermodynamic substrate and the macro-level monetary form.

This gives the conservation identity $\mu L = p^\top(I - A)x$ its full interpretation: the monetary value of the net product equals the social accounting of the total irreversible metabolic expenditure required to produce it. The first law holds in the circulation sphere (value conserved in exchange); the second law operates in the production sphere (labour irreversibly expended); and the MELT bridges the two.

6.4 Living Labour as Thermodynamic Work (*Lebendige Arbeit*)

Prigogine’s concept of a *dissipative structure* is sometimes invoked here — a system far from equilibrium maintained by continuous energy throughput. But this framing risks a

fundamental mischaracterisation: calling capitalism “dissipative” implies that labour is like *friction*, a process that degrades existing order into disorder. This is precisely backwards, and Marx’s vocabulary makes the correction exact.

In thermodynamics, there is a fundamental distinction between two modes of entropy production. *Dissipation* (friction, viscosity, electrical resistance) converts existing ordered energy into disordered heat: nothing is produced, energy is merely degraded, the output is waste. *Work* ($W = \int F \cdot dx$) is the directed expenditure of energy that accomplishes something — lifts a weight, drives a reaction, builds structure. The entropy produced in doing work is the unavoidable second-law *price* of that work (the Carnot inefficiency), not the work itself. A brake pad dissipates. A piston does work.

Labour is work, not dissipation. A worker does not degrade existing order like a brake pad; the worker transforms disordered raw materials into ordered commodities, adds new value, and extends the social division of labour. The entropy produced — metabolic heat, waste materials, depleted resources — is the thermodynamic *cost* of this work, the exhaust of the productive process. The commodity is the *output*; the entropy is the *exhaust*.

Marx names this distinction with precision. In *Capital* Vol. I Ch. 7, the labour process is *zweckmäßige Tätigkeit* — purposive activity, directed force. In the *Grundrisse*, Marx describes **lebendige Arbeit** (living labour) explicitly as fire: “living labour must seize upon these things and rouse them from their death-sleep, change them from merely possible use-values into real and effective ones.” Living labour is the fire that melts ore into steel, not the rust that corrodes it. It *animates* dead labour — the past labour congealed in capital goods and raw materials — by setting it in motion, adding new value through the work process. Dead labour is the fuel; living labour is the flame; the commodity is what the flame forges.

The thermodynamic structure of production, read through this distinction, runs as follows:

- **Free energy input:** food, fuel, raw materials — low-entropy resources accumulated over geological and biological time
- **Living labour** (*lebendige Arbeit*, thermodynamic work): the directed, purposive expenditure of human metabolic energy doing work on nature
- **Product** (local order): the commodity — organised useful structure, the objectification (*Vergegenständlichung*) of living labour in material form
- **Entropy output** (thermodynamic cost): metabolic heat, waste, depleted inputs — the unavoidable second-law exhaust of the work process
- **Internal dynamics:** the Hamiltonian redistribution of the value created — the approximately conservative circulation of already-produced value within the production network

But the *organizing principle* of this work process is not labour itself — labour does not organize its own exploitation. The organizing principle is capital: the profit-driven appropriation of living labour’s thermodynamic work through the exploitation relation. The MELT makes this precise. The equation $\mu = w(1 + \varepsilon)$ — where $\varepsilon = \mu/w - 1$ is the exploitation rate — is not merely an accounting identity; it is the social relation inscribed in the monetary

form. Capital’s claim on the value produced above the wage is the claim on the thermodynamic work performed above the worker’s own reproduction. Surplus value is surplus thermodynamic work, monetised through the MELT.

It is this profit motive — not labour’s thermodynamic character as such — that drives the ever-expanding accumulation of capital, the rising organic composition $\kappa = C/l$, and ultimately the metabolic rift. A society that organized living labour’s thermodynamic work without the profit motive would still produce entropy; it would not necessarily exhaust its entropy sinks. What makes the metabolic rift *structurally necessary* rather than contingent is the competitive pressure to accumulate: capital must reinvest surplus value, raise the scale of production, replace living labour with dead labour (machinery), and drive κ upward even as r^* falls. The second law asserts itself against the *capitalist* organization of the work process, not against work per se.

Foley and Smith (2007) identify the competitive equilibrium as the economic analogue of the maximum-entropy state in statistical mechanics: competition equalises profit rates across sectors, which is equivalent to maximising the number of ways a given total value can be distributed subject to the reproduction constraint. The equilibrium is not an energy minimum (as Walras supposed) but an entropy maximum — reached by the first-law redistribution dynamics of Sections 5 and 7. Labour is what sets the *level* of the value total that competition then distributes; the exploitation ratio ε is what determines how that total is divided between wages and profit, and hence how fast capital accumulates.

Mirowski demanded a genuine conservation law. The classical tradition delivers one — but thermodynamic, not mechanical. The first-law invariant is μL (total value created by living labour’s work on nature); the second-law constraint is that every increment requires the irreversible expenditure of thermodynamic work by living labour, at the necessary entropic cost; and the exploitation ratio ε is the social mechanism by which capital captures that increment and converts it into accumulation. Together they define the thermodynamic structure of capitalist social reproduction.

6.5 The Ecological Dimension

The second law at the production boundary takes the form of ecological limits. Capitalism has historically treated nature as a *free entropy sink* — a place to externalise the thermodynamic costs of production without charge. Pollution, climate change, soil exhaustion, biodiversity loss: these are the accumulating entropy of the capitalist production process dumped into natural systems without return.

Ricardian ground rent is the economic signal that this sink is becoming constrained. When natural resources are non-reproducible and scarce, their prices exceed their prices of production — the rent premium in Işıkara and Mokre’s (EXIOBASE) price-value deviation data for resource sectors. This rent is not value created by labour; it is the economic expression of a thermodynamic constraint: the cost of using a finite entropy sink. The growing ecological rent premium — visible in the systematic positive deviation of energy and mining

sector prices from their labour values — is the price system’s registration of the second law asserting itself at the boundary of the production network.

Saito (2020) and Moore (2011) articulate this as the metabolic rift: capitalism’s disruption of the natural cycles that sustain the conditions of production. From the thermodynamic perspective, the metabolic rift is the accumulation of unreturned entropy at the system’s natural boundary — the point at which the profit-driven accumulation of capital begins to undermine the natural entropy sink on which its exploitation of living labour depends.

7 Expanded Reproduction: Non-Conservative Dynamics

The simple reproduction case is a conceptual baseline, not a description of capitalism. Capitalism is defined by accumulation: the conversion of surplus value into additional capital, the expansion of the scale of production, and the continuous deepening of the social division of labour. Under expanded reproduction, the three conditions for Hamiltonian conservation are systematically violated. The violation is productive: the Hamiltonian level rises as living labour creates new value each period. But this expansion occurs within a nested dynamic structure whose long-run trajectory is governed by secular tendencies that Shaikh’s macro framework identifies.

7.1 Labour as Source of Value

In expanded reproduction, a portion of surplus value is reinvested. The production scale grows: $K(t+1) > K(t)$, $L(t+1) > L(t)$. Each period, new labour creates new value:

$$p(t)^\top (I - A) x(t) = \mu(t) L(t), \quad \frac{d}{dt}[\mu L] > 0.$$

The Hamiltonian is no longer conserved: $\dot{H} > 0$. Living labour — the second-law source term — continuously injects new thermodynamic work. The Hamiltonian level rises not because value is redistributed but because it has been irreversibly produced; the conservative manifold expands.

When expansion reverses — when credit contracts, investment collapses, or productive capacity is destroyed — the Hamiltonian falls: $\dot{H} < 0$. Capital is written off, workers are expelled from the circuit, anticipated values go unrealised. Unlike the conservative Goodwin redistribution, which preserves the total while oscillating around it, crisis destroys value: it reduces μL itself. The entropy within the economic structure increases, and the local order of the production network decays toward a lower level. The symmetric simple-reproduction regime is broken in both directions.

7.2 Non-Conservative Dynamics: The Unified $\dot{\mathcal{H}}$ Formula

When expanded reproduction prevails ($p > 0$, $\dot{K} > 0$), living labour injects new value into each model and the Hamiltonian is no longer conserved. Computing $\dot{H}$ across all three conservative frameworks yields a unified result.

Goodwin. In simple reproduction $\dot{H} = 0$ exactly. Net investment at rate $s_c r$ adds a source term $\Delta_q = s_c(r - r^*)$ to the employment-rate equation. The Hamiltonian drifts:

$$\dot{H}^{\text{Goodwin}} = \frac{\partial H}{\partial q} \Delta_q = (c - de^q) \cdot s_c(r - r^*)$$

which averages to $s_c S_\ell$ over a full cycle by the MELT identity $\Pi = \mu S_\ell$.

Flaschel–Semmler. Adding net investment $g = s_c r$ to the quantity equations ($\dot{z}_i \rightarrow \dot{z}_i + g$ for each sector i):

$$\dot{H}^{FS} = \sum_i \frac{\partial H}{\partial z_i} \cdot g = g \delta_p \mathbf{1}^\top M x = s_c S_\ell^{\text{aggregate}}$$

where $\mathbf{1}^\top M x = \mathbf{1}^\top (B - RA)x$ and the MELT converts net output in excess of the maximum-profit benchmark to surplus labour time.

Foley circuit. Derived directly from the stock equations in Section 5:

$$\dot{\mathcal{H}} = p S_\ell = \frac{p \varepsilon}{1 + \varepsilon} L$$

The result is the same across all three models:

Model	$\dot{H}$ — simple reproduction	$\dot{H}$ — expanded reproduction
Goodwin	0	$s_c \varepsilon L / (1 + \varepsilon)$
Flaschel–Semmler	0	$s_c \varepsilon L / (1 + \varepsilon)$
Foley circuit	0	$p \varepsilon L / (1 + \varepsilon)$

with $p = s_c$ by the Cambridge equation ($g = s_c r = pr$). **The time derivative of the Hamiltonian is uniquely determined by the surplus labour time and the reinvestment rate, regardless of which dynamic representation is used.** This is not a coincidence but a theorem: all three models describe different facets of the same underlying circuit, and the MELT is the invariant that bridges them. The circuit Hamiltonian $\mathcal{H} = K/\mu$ is capital in labour-time space; its time derivative $\dot{\mathcal{H}} = s_c S_\ell$ is the rate at which living labour time is being converted into accumulated capital — economic power in the thermodynamic sense.

Two limiting cases hold across all three. When $\varepsilon \rightarrow 0$: $\dot{H} = 0$ regardless of s_c — no surplus labour, no accumulation, simple reproduction is the only possibility. When $\varepsilon > 0$ and $s_c \rightarrow 1$ (all surplus reinvested): $\dot{H}$ is maximised — the full thermodynamic driving force of living labour converts into capital expansion. The exploitation rate ε is therefore not only a distributional parameter but the **thermodynamic intensity** of expanded reproduction: how forcefully the second law (living labour as irreversible work) drives the first-law conservative manifold outward.

7.3 Shaikh’s Macro Model: Three Nested Dynamics

Shaikh’s *Capitalism* (2016) offers the most systematic macro framework for capitalist dynamics on classical foundations. Rather than positing equilibrium and studying departures from it, Shaikh models capitalism as a system in permanent turbulent motion whose centre of gravity is itself moving. The framework layers three nested dynamic processes operating at distinct time horizons, each generating a distinct pattern of Hamiltonian behaviour (Shaikh 2016, 1992).

The Goodwin layer: turbulent equalization (3–8 years). At the shortest horizon, the conservative Goodwin dynamics provide the structural skeleton. But in Shaikh’s version, competition is “turbulent”: classical competition in Smith’s and Marx’s sense is active profit-seeking through cost reduction and price-cutting, not frictionless price-taking. The regulating capital in each sector — the producer operating at the best available conditions — sets the floor price; non-regulating capitals compete by investing in better conditions or accepting below-average returns. Profit differentials attract investment flows; over-investment drives down sectoral profitability; capital migrates until profit rates equalize in a turbulent, oscillating process — “turbulent gravitation” — never converging to rest.

In Hamiltonian terms, turbulent competition means that Δ_p and Δ_x are never exactly proportional to the identity. The Hamiltonian oscillates around its mean rather than being strictly conserved: approximately $\dot{H} \approx 0$, with small dissipative fluctuations from competitive unevenness. The Goodwin orbit is the dominant motion; the turbulence is its envelope.

The investment cycle: medium-run dynamics (7–11 years). At the medium horizon, Shaikh identifies the Juglar fixed-capital cycle. Firms invest in lumps — new plant, machinery, and infrastructure in waves as innovations spread and capacity gaps emerge — creating a systematic lag between profit signals and capacity response. The investment-profit interaction generates a two-variable oscillation. Let g_K be the growth rate of the capital stock and r the realized profit rate:

$$\dot{g}_K = \theta (r - r^*), \quad \dot{r} = -\sigma g_K + \text{const},$$

where θ governs how quickly investment responds to above-normal profitability and σ captures how quickly expanding capacity compresses profit rates through competition. The

system oscillates in (r, g_K) space around the long-period position (r^*, δ) where δ is the depreciation rate. These oscillations propagate through the input-output network via the off-diagonal dynamics of the Flaschel–Semmler framework: the adjustment speed matrices Δ_x vary with investment lags, and the resulting waves of capital goods demand, then consumer goods demand, then labour demand constitute the Juglar rhythm.

In Hamiltonian terms: during expansion, $\dot{H} > 0$ — new investment mobilises new living labour, which creates new value; the conservative manifold expands. During contraction, excess capacity yields losses and partial capital destruction: $\dot{H} < 0$, as anticipated value goes unrealised. Unlike the Foley financial cycle (which is conservative on average), the Juglar involves *real* capital creation and destruction — genuine non-conservation at the medium frequency, bounded above and below by the investment-profit dynamic.

The long wave: secular dynamics and the falling rate of profit (\$ \$50 years). At the longest horizon, Shaikh (1992) identifies the Kondratiev long wave, driven by the secular tendency of the rate of profit to fall. The basic rate of profit is:

$$r^* = \frac{\varepsilon}{1 + \varepsilon} \cdot \frac{1}{\kappa}$$

where ε is the exploitation rate and $\kappa = C/l$ is the materialised organic composition of capital. Even as ε rises with intensification and deskilling, $\varepsilon/(1 + \varepsilon)$ is bounded above by 1. The dominant long-run force is κ , whose evolution is governed by:

$$\dot{\kappa} = (g_K - g_L - \delta) \kappa,$$

where g_L is the growth rate of living labour. **Marx-biased technical change** — the competitive pressure to replace living labour with machinery in order to cut unit costs — systematically drives $g_K > g_L + \delta$, so $\dot{\kappa} > 0$ and:

$$\frac{dr^*}{d\kappa} = -\frac{\varepsilon}{(1 + \varepsilon) \kappa^2} < 0.$$

As $\kappa \rightarrow \infty$, $r^* \rightarrow 0$: the ratio of living to dead labour falls, and with it the capacity of the system to generate surplus. This is the *internal* source of secular decline — the exhaustion of living labour’s relative weight in production, developed by Marx.

A second, *external* source operates via ecological ground rent (developed by Ricardo): the depletion of natural resources and the saturation of the entropy sink raises rents, diverting an increasing share of gross profit to resource owners and compressing the net profit rate further. The two forces are formally additive: $r_{\text{net}}^* = r_{\text{Marx}}^* - \phi(\text{rent})$, where ϕ rises as natural limits bind.

Together they define the thermodynamic trajectory of capitalism toward what might be called its **secular entropy maximum**: as living labour is displaced by dead labour (Marxian force) and as natural entropy sinks are exhausted (Ricardian force), the surplus available

for accumulation tends to zero — not instantaneously, but as a structural limit approached asymptotically. Each long-wave trough involves crisis-driven devaluation that resets κ downward and temporarily restores r^* , but the secular envelope continues its descent.

The growth rate of profitable accumulation takes the form:

$$g_P^* = -a + s_c \cdot r^*$$

where s_c is the capitalists' savings (investment) propensity and a is the autonomous rate of technical change. Profitable accumulation requires $s_c \cdot r^* > a$; the long wave is the oscillation of this condition as r^* moves around its secular trend. The trough — Marx's "point of absolute overaccumulation" — is reached when $r^{**} = a/s_c$ and $g_P^* = 0$. Recovery requires sufficient devaluation to restore r^* above r^{**} .

The three layers are not independent but interpenetrating: the long wave modulates the amplitude of the Goodwin orbit; the investment cycle modulates its frequency; the Goodwin cycle describes the orbit itself. Shaikh also incorporates a reserve army mechanism: a rising profit rate drives higher accumulation, absorbs the reserve army of labour, raises wages, and compresses profits — the Goodwin mechanism operating at long-wave frequency, superimposed on the Kondratiev cycle.

7.4 The Long Wave as Evolution of the Conservative Manifold

The Hamiltonian interpretation of Shaikh's framework makes precise what is implicit in his analysis.

The Goodwin layer is approximately conservative: $\dot{H} \approx 0$. The wage-profit orbit preserves total value while redistributing it between classes — the first-law dynamics within a given production regime.

The investment-cycle layer introduces systematic drifts: $\dot{H} > 0$ in expansion, $\dot{H} < 0$ in contraction. The drifts are bounded and cyclical; the medium-run Hamiltonian oscillates around a slowly moving mean.

The long-wave layer introduces the secular envelope. The maximum rate of profit $R = 1/\lambda_1 - 1$ declines as accumulated capital raises the dominant eigenvalue λ_1 of the production matrix A . The conservative manifold itself contracts: even as individual Goodwin orbits maintain their approximate internal conservation, the equilibrium around which they orbit shifts as the organic composition rises. The amplitude of sustainable redistribution — the size of the conservative orbit consistent with positive profitability — shrinks as R falls. As the prices-of-production formula shows, p^* becomes singular as $r \rightarrow R$: the range of prices consistent with viable production narrows as the maximum rate of profit declines.

The long wave is therefore not a departure from conservative dynamics but the secular evolution of the conservative manifold itself. Individual Goodwin cycles remain approximately

conservative within each production regime; the regimes evolve under the pressure of accumulation; and the sequence of regimes traces a slow contraction of the space of possible conservative orbits. Crisis recovery — the long-wave trough — is the event that resets the manifold: sufficient destruction of fixed capital lowers λ_1 , raises R , and creates space for a new expansion. The reset is a discontinuous event: concentrated devaluation of the capital stock restores profitability and relaunches the conservative dynamics on a new manifold.

8 Conclusion

The argument of this paper can be stated in two levels.

At the *thermodynamic level*, production involves both laws. The first-law analogue (Marxian conservation): the total monetary value of the net product equals MELT times living labour for any price system — Shaikh’s universal result ($\Pi = S$ unconditionally). The second-law analogue (labour as irreversible process): abstract labour is the economic expression of metabolic irreversibility; the MELT converts this thermodynamic quantity to money; the economy is a dissipative structure sustained by continuous labour throughput. Mirowski demanded a genuine conservation law; the classical tradition delivers two, operating at different levels — and the second law is what makes the first possible, by ensuring that the “energy” entering the system (new value created by labour) is genuinely produced and irreversibly consumed, not conjured from a tautological accounting identity.

At the *dynamical level*, the Hamiltonian of Goodwin and Flaschel–Semmler captures the first-law redistribution dynamics: competitive circulation is frictionless in the ideal case, with profit-rate equalisation and wage-profit oscillation tracing conservative orbits. The Hamiltonian is conditionally conserved — within a period, given the technique, in the absence of structural asymmetry. It drifts upward in expanded reproduction (as new labour creates new value), downward in crisis (as capital is devalued), and secularly downward over the long wave (as Shaikh’s falling rate of profit erodes the equilibrium). The secular decline has two thermodynamic sources: the **Marxian** source — Marx-biased technical change raises the organic composition κ , displacing living labour with dead labour and driving $r^* \rightarrow 0$ as $\kappa \rightarrow \infty$; and the **Ricardian** source — the exhaustion of natural entropy sinks raises ground rent, compressing net profitability from outside. Together they trace the system’s asymptotic approach to a *secular entropy maximum*: the point at which the surplus available for accumulation vanishes, not through any single catastrophe but as the structural limit of a system that simultaneously displaces its own source of value (living labour) and fills its own entropy sink (nature).

Mirowski’s mistake was to demand a *mechanics* analogy: a closed, self-sustaining, time-reversible system. The correct analogy was always thermodynamics. But rather than simply substituting thermodynamics for mechanics, the classical tradition — when fully reconstructed — shows that both are needed: mechanics for the circulation dynamics (first law,

Hamiltonian), thermodynamics for the production process (second law, irreversibility), and the ecological crisis as the point where the second law asserts itself at the natural boundary of the system.

The ongoing debates in heterodox political economy — Işıkara and Mokre on international value transfers, Rotta on complexity rents, Saito on metabolic rift — are not peripheral extensions of this framework but its necessary completion. International trade introduces persistent non-conservation at the world level (differential organic compositions break the universal conservation identity across national boundaries). Ecological scarcity introduces the second law at the production boundary (ground rent as the price of a finite entropy sink). And financialisation introduces spurious first-law violations in the circulation sphere (credit money creation as nominal value not backed by labour). Each is a distinct mode by which the thermodynamic structure of social reproduction is breached. Together they constitute the research programme that a fully thermodynamic political economy requires.

9 References

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